Brijodison

Engineering Notebook Element L

Engineering Service Learning

Presentation of Designer’s Recommendations

If another person was going into this same project in the future, there are many pieces of advice I would give them. These include having an open mind, focusing on simplicity, and the implementation of a compressible medium. We eventually realized these as the project went on, but it could’ve been a lot smoother if we had known any of these factors earlier. All of these pieces of advice can be applied to similar situations as well.

The first piece of advice is making sure to keep an open mind. At the beginning of the prototyping phase, we came up with almost 30 designs. We narrowed it down to 6 designs and I felt that my designs were some of the better options. Based on our criteria and constraints, we created a design matrix that weighted the effectiveness of each design. To my surprise, some of my designs were the worst ones based on our matrix. We ended up agreeing on a design and after we chose it, we still ended up pivoting and changing it as we went on. After that, we realized that the plan we had laid out wasn’t perfect and there would be many changes throughout the process. As long as we kept an open mind we knew we would be able adapt after each problem we faced.

The biggest problem we ran into was over complicating our designs. After we did our decision matrix the main reason for certain designs not being chosen was the amount of complexity each design had. I was hyper focused on the idea of an adjustable cart handle at first that had multiple moving parts; we realized that a cart handle that wasn’t adjustable, but rather had a range of heights to grab onto would be better and leave far less room for error. Another reason we wanted to keep the design simple was the specific criteria and constraints of this project, the main one being the budget. The budget for this project is $30; with this budget we couldn’t afford a failed attempt while constructing this. By keeping it simple, we could make the cart attachment easily attachable and detachable. Not only can it be quickly taken on/off, but also requires no bending over, which is another very important requirement. If we would’ve gone for a design that had multiple moving pieces, one part not working means the entire device is rendered useless.

The final piece of advice we would give is to implement the use of a compressible medium. Going into the prototyping phase, we took measurements of the cart and cut our device to fit the cart perfectly. While this seemed effective in our mind, we never took friction into consideration. After we had already made our first prototype it slid way too much, and completely starting over didn’t seem smart since we still felt good about our design. By adding a compressible medium in rubber we nearly eliminated the friction along the bottom of the cart, as well as where the PVC poles meet the top and bottom of the handle. A compressible medium can be used in any engineering situation where something may not be a perfect fit, or even where something is sliding too much, not only in this instance.

These 3 pieces of advice would be valuable to anyone who is taking on this project or even a group project similar to this one. Keeping an open mind is important to be able to adapt to challenges you run into. Keeping it simple will reduce room for error and allow more time for a project’s revisions. Finally, using a compressible medium is important in any project for anything that doesn’t perfectly fit in one another.